

Bruno do Rosario Petrucci

PhD candidate

Ecology, Evolution, and Organismal Biology department
Iowa State University

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brpetrucci.github.io

Education

- 2020–2027 **PhD in Ecology and Evolutionary Biology**, Iowa State University, Ames, IA
Thesis: “Evaluating the performance of fossil-aware phylogenetic methods, with an application to Canidae”
- 2016–2020 **B.S. in Computational and Applied Mathematics with Honors**, University of Chicago, Chicago, IL
Magna Cum Laude (2020), Dean’s List (2017–2020)

Research

- 2020–2027 **Department of Ecology, Evolution, and Organismal Biology**, Iowa State University, Ames, IA, PI: Tracy Heath
Integrating fossils into phylogenetic inference and comparative methods, with a focus on Canidae.
- 2019–2020 **Department of Geophysical Sciences**, University of Chicago, Chicago, IL, PI: Michael Foote
Statistical modeling of fossil occurrence data.
- 2019 **Department of Ecology**, University of São Paulo, São Paulo, Brazil, PI: Tiago Quental
Simulations of phylogenetic trees and fossil records (paleobuddy).
- 2018–2019 **Department of Ecology & Evolution**, University of Chicago, Chicago, IL, PI: Gregory Dwyer
Eco-evolutionary modeling of fungal diseases in spongy moths.
- 2017–2018 **Kavli Institute for Cosmological Physics**, University of Chicago, Chicago, IL, PI: Luca Grandi
Simulations of particle collisions for dark matter detector calibrations.

Publications

- 2025 **B. do Rosario Petrucci**, M.R. May, T.A. Heath. “Fossils improve extinction-rate estimates under state-dependent diversification models.” *Philosophical Transactions of the Royal Society B* 380, 20230313. DOI: 10.1098/rstb.2023.0313.
- 2022 **B. do Rosario Petrucci**, M. Januario, T.B. Quental. “paleobuddy: An R package for flexible simulations of species diversification and sampling.” *Methods in Ecology and Evolution* 13(12):2692–2698. DOI: 10.1111/2041-210X.13996.
- 2022 Barido-Sottani, J., J.A. Justison, R. Borges, J.M. Brown, W. Dismukes, **B. do Rosario Petrucci**, L. Guimarães Fabreti, S. Höhna, M.J. Landis, P.O. Lewis, M.R. May, F.K. Mendes, W. Pett, B.D. Redelings, C.M. Tribble, A.M. Wright, R. Zenil-Ferguson, and T.A. Heath. “Lessons learned from organizing and teaching virtual phylogenetics workshops.” *The Bulletin of the Society of Systematic Biologists* 1(2):8425. DOI: 10.18061/bssb.v1i2.8425.

Manuscripts Submitted or In Preparation

‡ indicates equal contribution.

- In prep **B. do Rosario Petrucci**, G.J. Slater, S. Sharma, U. Stolz, M. Januário, T.A. Heath. “Evaluating the accuracy of the stratigraphic-range fossilized birth-death process to estimate the phylogeny of extinct and living Canidae (Mammalia: Carnivora).”
- In prep R. Zenil-Ferguson‡, **B. do Rosario Petrucci**‡, N. Castillo-Rodríguez, S. Brock, T.A. Heath. “A Bayesian approach to state-dependent diversification models.”
- In prep I. Novick, T. Wells-Green, **B. do Rosario Petrucci**, T. Cheung, E. Ponce, H. E. Aichelman, S. Mullen. “Morphological vs. molecular identification methods of synanthropic microlepidoptera.”

In prep I. Novick, H. E. Aichelman, **B. do Rosario Petrucci**, J. D. Alqassar, C. Lin, K. Chen, S. W. Davies, M. D. Sorenson, S. P. Mullen. "Population genomics of the webbing clothes moth (*Tinea bisselliella*) reveals cryptic structure with implications for pest management."

Presentations

Organized Symposia

2026 *Cross-Disciplinary Approaches for Understanding Budding Speciation*, Virtual Evolution meeting.

Invited Talks

2026 *Budding phylogenetic trees: inference with BEAST2 and post-processing with paleobuddy*. Palaeoverse lecture series.

2026 *Advances towards tree distance measures for budding phylogenetic trees*. Virtual Evolution meeting (Cross-Disciplinary Approaches for Understanding Budding Speciation symposium).

2024 *State-dependent diversification with fossils: a case study with Canidae*. Evolution meeting (talk), ICERM Algorithmic Advances and Implementation Challenges: Developing Practical Tools for Phylogenetic Inference workshop (poster).

2019 *PaleoBuddy: An R Package for simulation of fossil records and phylogenies with variable rates and age dependency*. Paleontology Club at the University of Chicago.

Contributed Talks

2026 *Evaluating the accuracy of the stratigraphic-range fossilized birth-death process to estimate the phylogeny of extinct and living Canidae*. Evolution meeting.

2025, 2026 *Inferring a complete phylogenetic tree of Canidae under the stratigraphic-range fossilized birth-death process*. Evolution meeting (2025), Society for Systematic Biologists meeting (2026).

2023 *Accuracy of extinction rate estimates using state-dependent diversification models with fossil data*. Evolution meeting, Great Lakes Student Paleoconference.

2021 *Accuracy of diversification rate inference methods for shifts in trait-dependence*. Virtual Evolution meeting.

2020 *PaleoBuddy: Simulating evolutionary dynamics in R*. National Collegiate Research Conference (poster), University of Chicago Physical Sciences Division Undergraduate Research Symposium.

Software

paleobuddy **Flexible simulation of phylogenetic trees, fossil records, and character data.**
(R) Developer and maintainer.  github repo.

RevBayes **Bayesian phylogenetic inference with the use of graphical models relying on an interpreted programming language interface.**
(C++) Contributor.  github repo.

Teaching

2021–2026 **Principles of Biology Laboratory I**, *Iowa State University*, Ames, IA, Instructor
9 semesters.

2024, 2025 **Workshop on Molecular Evolution**, *Marine Biological Laboratory*, Woods Hole, MA, Teaching assistant

2021, 2022 **Biological Evolution**, *Iowa State University*, Ames, IA, Instructor
Summer course.

2021 **RevBayes Stay-At-Home Workshop**, *Virtual*, Teaching assistant

2019 **Introduction to Quantitative Modelling in Biology (Basic)**, *University of Chicago*, Chicago, IL, Teaching assistant

- 2018–2019 **Elementary Functions and Calculus I-II**, *University of Chicago*, Chicago, IL, Junior Tutor
- 2017 **Calculus I**, *University of Chicago*, Chicago, IL, Teaching assistant
- 2017 **Mathematical Methods for the Physical Sciences I**, *University of Chicago*, Chicago, IL, Grader
- 2015 **Physics, Chemistry, and Biology Lab**, *Colégio Vital Brazil*, São Paulo, Brazil, Lab Assistant

Workshops

- 2026 **Academic Job Market Boot Camp**, *Beyond the Professoriate*, Online, Participant
- 2026 **Project Management Skills for Emerging Researchers**, *Iowa State University*, Ames, IA, Participant
- 2024 **Algorithmic Advances and Implementation Challenges: Developing Practical Tools for Phylogenetic Inference**, *Brown University (ICERM)*, Providence, RI, Participant/poster presenter
- 2024 **NSF Pathways to Open Source Ecosystem**, *I-Corps Program*, Online, Technical lead (RevBayes team)
- 2023 **RevBayes Hackathon**, *Paleontological Museum*, Munich, Germany, Contributor
- 2022 **Workshop on Molecular Evolution**, *Marine Biological Laboratory*, Woods Hole, MA, Participant

Professional Service

Graduate Research in Evolutionary Biology and Ecology (Iowa State University)

- 2023–2024 Social Chair
- 2022–2023 President
- 2021–2022 Social Chair

EEOB Department Graduate Student Organization (Iowa State University)

- 2025–2026 Empowerment, Community, and Opportunity Committee Representative
- 2023–2024 President
- 2022–2023 Facilities, Graduate Studies, and Honors & Awards Committees Representative
- 2021–2022 Social Chair

Outreach

- 03/26/2026 **Biological Sciences Undergraduate Information Panel**, *69th State Science & Technology Fair of Iowa*, Ames, IA, Invited Panelist
- 02/26/2025 **Graduate School Advice Panel**, *Iowa State University SACNAS and STEM Scholars program*, Ames, IA, Invited Panelist
- 05/2024 **EEOB GSO Annual Plant Sale**, *Reiman Gardens*, Ames, IA, Organizer
- 01/23/2023 **Annual Ecology and Evolutionary Biology Symposium**, *Iowa State University*, Ames, IA, Organizer
- 2021 **Iowa Mentored Community College Teaching Practicum**, *Des Moines Area Community College*, Boone, IA, Mentee/teaching assistant

Awards & Honors

- 2025 EEOB Teaching Award
- 2024 Iowa State University Teaching Excellence Award
- 2024 EEOB Service Award
- 2023 EEOB Teaching Award (Honorable Mention)
- 2023 Methods in Ecology and Evolution's Robert May Prize Award (Shortlist)

Fellowships & Grants

- 2026–2027 **Brown Graduate Fellowship**, *Iowa State University*, \$10,000
- 2025–2026 **EEOB GSO Travel Award**, *Iowa State University*, \$500
3 semesters
- 2022–2025 **EEB Professional Development Grant**, *Iowa State University*, \$400
4 semesters
- 2024 **ICERM Travel Grant**, *Brown University*, around \$2,000)
- 2020–2021 **EEB Fellowship**, *Iowa State University*, \$9,600
2 semesters
- 2020–2021 **LAS Graduate Scholar Award**, *Iowa State University*, \$3000
- 2018–2020 **College Research Fellows Program**, *University of Chicago*, \$15/h salary
5 quarters
- 2019 **Jeff Metcalf Global Fellowship Grant**, *University of Chicago*, \$5000
- 2018 **UCISTEM Summer Research Grant**, *University of Chicago*, \$1000

Society Membership

- 2020–Current Society of Systematic Biologists, graduate student member